Pratt Coordinates for Natural Numbers and Monic Rational Polynomials

Reconstruction, Hilbert-Space Embeddings, and a Pratt-Coordinate Form of Mason–Stothers

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Status of this document. This is a living research note intended to be revised and enlarged. Version 0.2 contains the foundational reconstruction and Hilbert-space theory from Version 0.1 and adds a complete Pratt-coordinate proof of the Mason–Stothers theorem, a computational test of the proposed replacement $M_x \rightsquigarrow M_\varphi$, a counterexample to the naive coordinatewise extension, a valid capacity bound for every Pratt coordinate, and a first discussion of the addition defect that obstructs a direct transfer to the integer *abc* problem.

Abstract

We compare positive integers with monic polynomials in $\mathbb{Q}[x]$ through recursive factorization trees. For a positive integer n, the Pratt exponent $m_p(n)$ counts occurrences of the prime p in the Pratt forest obtained by recursively factoring $q-1$ below every prime node q. For a monic polynomial $f \in \mathbb{Q}[x]$, we define analogous exponents $M_\varphi(f)$ indexed by monic irreducible polynomials and prove the finite reconstruction formula

$$f = \prod_{\varphi} \left(\frac{\varphi}{D\varphi} \right)^{M_\varphi(f)}. \tag{3}$$

For the recursively defined family $f_n(x)$, the prime-index polynomials f_p are irreducible, $M_{f_p}(f_n) = m_p(n)$, and formula (3) specializes first to a polynomial product formula and then, at $x = 2$, to the numerical Pratt product formula. These coordinates give injective multiplicative-to-additive embeddings of monic rational polynomials and positive integers into Hilbert spaces.

The second part applies reconstruction to Mason–Stothers. A Pratt boundary operator recovers the ordinary irreducible valuations from the recursive coordinates. Logarithmic differentiation then yields $f/\operatorname{rad}_P(f) \mid f'$, and a Wronskian argument proves Mason–Stothers entirely in Pratt language. The degree is exactly the terminal coordinate $M_x(f)$. Computation and explicit counterexamples show that replacing M_x by an arbitrary coordinate M_ψ in the same inequality is false. We prove instead a universal capacity bound

$$M_x(\psi) M_\psi(f) \leq M_x(f),$$

which produces a valid Mason bound for every coordinate. Finally, specialization to f_n gives unconditional inequalities for numerical Pratt coordinates and isolates the addition defect $f_a + f_b - f_{a+b}$ as the main obstruction to an integer *abc* application.

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Part I

Foundations: Reconstruction and Hilbert-Space Coordinates

1 Introduction and guiding analogy

Unique factorization gives two familiar coordinate systems:

$$n = \prod_{p\in\mathbb{P}} p^{v_p(n)}, \qquad f = \prod_{\varphi\in\mathcal{I}} \varphi^{v_\varphi(f)}.$$

Here $\mathbb{P}$ is the set of primes and $\mathcal{I}$ is the set of nonconstant monic irreducible polynomials in $\mathbb{Q}[x]$. The ordinary exponents $v_p(n)$ and $v_\varphi(f)$ record only the top-level factorization. Pratt coordinates retain the factorization of the ancestors as well: below a prime q one factors $q-1$, and below an irreducible polynomial φ one factors $D\varphi$.

The central chain of implications in this note is

$$\boxed{\begin{array}{c}\text{general polynomial reconstruction (3)}\\ \Downarrow \\ \text{specialized polynomial formula (2)}\\ \Downarrow \\ \text{integer formula (1)}\end{array}}$$

The first formula is the structural theorem. The other two are specializations generated by the family f_n and the evaluation map $f \mapsto f(2)$.

The polynomial construction should be viewed as an algebraic analogue rather than a literal copy. In the integer setting the distinguished terminal prime is 2, because $2-1=1$. In the polynomial setting the distinguished terminal irreducible is x, for which we impose $Dx = 1$. The local reconstruction weights are respectively

$$\frac{p}{p-1} \qquad \text{and} \qquad \frac{\varphi}{D\varphi}.$$

2 Pratt exponents for positive integers

2.1 Pratt trees and forests

Definition 2.1 (Pratt tree of a prime)**.** Let q be prime. Its Pratt tree T_q is defined recursively. The root is labelled q. If $q = 2$, the tree has no children. If $q > 2$ and

$$q - 1 = \prod_{r \in \mathbb{P}} r^{v_r(q-1)},$$

then the root has $v_r(q-1)$ children labelled r for every prime divisor r of $q-1$, and below each such child one places a copy of T_r.

Definition 2.2 (Pratt forest and Pratt exponent)**.** For

$$n = \prod_{q \in \mathbb{P}} q^{v_q(n)},$$

the Pratt forest of n is the disjoint union of $v_q(n)$ copies of T_q. For a prime p, the *Pratt exponent* $m_p(n)$ is the number of vertices labelled p in this forest.

Equivalently, the Pratt exponents are characterized recursively by

$$m_p(1) = 0, \tag{1}$$
$$m_p(ab) = m_p(a) + m_p(b), \tag{2}$$
$$m_p(q) = \mathbf{1}_{\{p=q\}} + m_p(q-1) \qquad (q \text{ prime}). \tag{3}$$

These rules are well-founded because every prime divisor of $q-1$ is smaller than q.

Example 2.3. The tree of 5 has one vertex labelled 5 and two vertices labelled 2, because $5 - 1 = 2^2$. Hence

$$m_5(5) = 1, \qquad m_2(5) = 2,$$

and all other Pratt exponents vanish.

2.2 The numerical reconstruction formula

The numerical formula to be recovered later is

$$\boxed{n = \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p}\right)^{-m_p(n)}.} \tag{1}$$

The product is finite because the Pratt forest of a fixed integer is finite. This formula is developed systematically in [1]. In this note it will appear as the final specialization of the polynomial reconstruction theorem.

3 The recursive polynomial family $f_n(x)$

The following family was introduced to reflect multiplication of integers inside $\mathbb{Z}[x]$; see [2] and the MathOverflow discussion [3].

Definition 3.1 (The family f_n)**.** Set

$$f_1(x) = 1, \qquad f_2(x) = x.$$

For every odd prime p, define recursively

$$f_p(x) = 1 + \prod_{q \mid p-1} f_q(x)^{v_q(p-1)}. \tag{4}$$

For an arbitrary positive integer

$$n = \prod_{p \in \mathbb{P}} p^{v_p(n)},$$

define

$$f_n(x) = \prod_{p \mid n} f_p(x)^{v_p(n)}. \tag{5}$$

The recursion is well-defined because every prime divisor of $p-1$ is smaller than p.

Proposition 3.2 (Elementary properties)**.** *For all $m, n \geq 1$:*

(i) $f_n \in \mathbb{Z}[x]$ *is monic and has nonnegative coefficients;*

(ii) $f_{mn} = f_m f_n$*;*

(iii) $f_n(2) = n$*;*

(iv) $f_n(0) = 0$ *if n is even, and $f_n(0) = 1$ if n is odd.*

Proof. The first statement follows by induction from (4) and multiplicativity. The second is immediate from additivity of the ordinary prime valuations:

$$v_p(mn) = v_p(m) + v_p(n).$$

For the third statement, first let p be odd and assume the claim already holds for every prime divisor of $p-1$. Then

$$f_p(2) = 1 + \prod_{q \mid p-1} q^{v_q(p-1)} = 1 + (p-1) = p.$$

The general case follows from (5). Finally, $f_2(0) = 0$. If p is odd, then $p-1$ is even, so f_{p-1} contains the factor $f_2 = x$ and has constant term zero; therefore $f_p(0) = 1$. Multiplicativity gives the parity statement for all n. $\square$

The first few prime-index polynomials are

p	$f_p(x)$
2	x
3	$x+1$
5	x^2+1
7	x^2+x+1
11	x^3+x+1
13	x^3+x^2+1
17	x^4+1
19	$x(x+1)^2+1.$

4 Irreducibility of the prime-index polynomials

We include a complete proof, following the root-location strategy due to Jonathan Love and the presentation in [3, 2]. The statement below corrects a harmless base-case issue by formulating the modulus bound only for odd primes.

Lemma 4.1 (A left-half-plane factor criterion). *Let $g \in \mathbb{Z}[x]$ be monic with $g(0) = \pm 1$. Suppose every complex root θ of g satisfies*

$$\mathrm{Re}(\theta) < -\frac{1}{2}.$$

Then every nonconstant irreducible factor of g in $\mathbb{Z}[x]$ is $x + 1$. Consequently, $g = (x + 1)^r$ for some $r \geq 0$.

Proof. Let h be a nonconstant monic irreducible factor of g. By Gauss's lemma, $h \in \mathbb{Z}[x]$ and $h(0) = \pm 1$. If $\theta_1, \ldots, \theta_d$ are the roots of h, then for each j,

$$|\theta_j + 1| < |\theta_j|,$$

because the point θ_j lies strictly to the left of the perpendicular bisector $\mathrm{Re}\, z = -1/2$ of 0 and -1. Hence

$$|h(-1)| = \prod_{j=1}^{d} |\theta_j + 1| < \prod_{j=1}^{d} |\theta_j| = |h(0)| = 1.$$

Since $h(-1)$ is an integer, $h(-1) = 0$. Thus $x + 1$ divides the irreducible polynomial h, and therefore $h = x + 1$. □

Lemma 4.2 (Root location). *Let p be an odd prime. If $z \in \mathbb{C}$ satisfies*

$$\mathrm{Re}(z) \geq \frac{3}{2},$$

then

$$|f_p(z)| > 2.$$

In particular, every root θ of f_p satisfies

$$\mathrm{Re}(\theta) < \frac{3}{2}.$$

Proof. We induct over the odd primes. For $p = 3$,

$$|f_3(z)| = |z + 1| \geq \mathrm{Re}(z) + 1 \geq \frac{5}{2} > 2.$$

For $p = 5$, writing $z = a + bi$ with $a \geq 3/2$, we have

$$\begin{aligned}
|z^2 + 1|^2 &= (a^2 - b^2 + 1)^2 + 4a^2b^2 \\
&= \big(a^2 + (b-1)^2\big)\big(a^2 + (b+1)^2\big) \\
&\geq a^4,
\end{aligned}$$

so $|f_5(z)| \geq a^2 \geq 9/4 > 2$.

Now let $p \geq 7$ and assume the assertion for all odd prime divisors of $p-1$. From the defining recursion and the reverse triangle inequality,

$$|f_p(z)| \geq \prod_{q|p-1} |f_q(z)|^{v_q(p-1)} - 1.$$

If $p-1$ has an odd prime divisor q, then the product contains a factor $|f_2(z)| = |z| \geq 3/2$ and a factor $|f_q(z)| > 2$; every remaining factor has modulus greater than 1. Therefore $|f_p(z)| > 3 - 1 = 2$.

If $p - 1 = 2^k$, then $k \geq 3$ and

$$f_p(z) = 1 + z^k.$$

Thus

$$|f_p(z)| \geq |z|^k - 1 \geq \left(\frac{3}{2}\right)^3 - 1 > 2.$$

A root in the half-plane $\mathrm{Re}(z) \geq 3/2$ would contradict the modulus bound. □

Theorem 4.3 (Irreducibility of f_p)**.** *For every prime p, the polynomial $f_p(x)$ is irreducible in $\mathbb{Z}[x]$, and hence in $\mathbb{Q}[x]$.*

Proof. The case $p = 2$ is immediate. Let p be odd and suppose that f_p is reducible. Since f_p is monic and primitive, Gauss's lemma gives a factorization

$$f_p(x) = u(x)v(x)$$

with nonconstant monic polynomials $u, v \in \mathbb{Z}[x]$. Evaluation at $x = 2$ gives

$$p = u(2)v(2).$$

Because p is prime, one factor has absolute value 1; after interchanging u and v, assume $u(2) = \pm 1$.

Set $g(x) = u(x+2)$. Then g is monic, $g(0) = \pm 1$, and every root η of g has the form $\eta = \theta - 2$, where θ is a root of f_p. By Theorem 4.2,

$$\mathrm{Re}(\eta) < -\frac{1}{2}.$$

By Theorem 4.1, $g = (x+1)^r$ for some $r \geq 1$, and hence

$$u(x) = (x-1)^r.$$

It follows that $u(1) = 0$ and therefore $f_p(1) = 0$. This is impossible because f_p has nonnegative coefficients and constant term 1. Thus f_p is irreducible. □

Corollary 4.4 (Prime detection)**.** *The polynomial f_n is irreducible in $\mathbb{Q}[x]$ if and only if n is prime.*

Proof. The prime case is Theorem 4.3. If n is composite, its prime factorization makes (5) a nontrivial product of nonconstant polynomials. □

Corollary 4.5 (Distinctness)**.** *If p and q are distinct primes, then $f_p \neq f_q$.*

Proof. By Theorem 3.2, $f_p(2) = p \neq q = f_q(2)$. □

5 The polynomial predecessor D and the local ratio Δ

Let

$$\mathcal{I} = \{\varphi \in \mathbb{Q}[x] : \varphi \text{ is nonconstant, monic, and irreducible}\}.$$

The role of the terminal prime 2 is played by x.

Definition 5.1 (Polynomial predecessor and local ratio)**.** For $\varphi \in \mathcal{I}$, define

$$D\varphi = \begin{cases} 1, & \varphi = x, \\ \varphi(x) - \varphi(0), & \varphi \neq x, \end{cases} \qquad \Delta\varphi = \frac{\varphi}{D\varphi} \in \mathbb{Q}(x)^{\times}. \tag{6}$$

The special convention $Dx = 1$ is essential. Without it the naive formula $Dx = x - x(0) = x$ would make the recursion loop forever at x.

Lemma 5.2 (Descent)**.** *For every $\varphi \in \mathcal{I}$, the polynomial $D\varphi$ is monic. If $\varphi \neq x$, then $x \mid D\varphi$. Moreover:*

- *if* $\deg \varphi = 1$ *and $\varphi \neq x$, then $D\varphi = x$;*
- *if* $\deg \varphi = d \geq 2$*, every irreducible factor of $D\varphi$ has degree strictly smaller than d.*

Proof. If $\varphi \neq x$ is irreducible, then $\varphi(0) \neq 0$; otherwise x would divide φ. Subtracting the constant term preserves the leading coefficient and makes the constant term zero, so $D\varphi$ is monic and divisible by x. For a monic linear polynomial $\varphi = x - a$ with $a \neq 0$, one gets $D\varphi = x$. If $d \geq 2$, the factor x consumes positive degree, so no irreducible factor can have degree d. □

This descent is the polynomial analogue of the numerical inequality $q < p$ for every prime divisor q of $p - 1$.

6 Polynomial Pratt exponents

6.1 Recursive definition

For a monic polynomial $h \in \mathbb{Q}[x]$, write its unique factorization as

$$h = \prod_{\eta \in \mathcal{I}} \eta^{v_\eta(h)},$$

where only finitely many exponents are nonzero. We include 1 as the empty product.

Definition 6.1 (Polynomial Pratt exponent on irreducibles)**.** For $\psi, \varphi \in \mathcal{I}$, define $M_\psi(\varphi)$ recursively by

$$M_\psi(\varphi) = \mathbf{1}_{\{\psi = \varphi\}} + \sum_{\eta \in \mathcal{I}} v_\eta(D\varphi) M_\psi(\eta). \tag{7}$$

For $\varphi = x$, the sum is empty because $Dx = 1$.

Proposition 6.2 (Well-definedness and finite support)**.** *The recursion (7) uniquely defines nonnegative integers $M_\psi(\varphi)$. For each fixed φ, only finitely many ψ have $M_\psi(\varphi) \neq 0$.*

Proof. Begin with x, for which
$$M_\psi(x) = \mathbf{1}_{\{\psi=x\}}.$$
Every other monic irreducible linear polynomial has predecessor x, so its exponents are then defined. Proceed by induction on degree. If $\deg\varphi = d \ge 2$, all irreducible factors of $D\varphi$ have degree smaller than d by Theorem 5.2; their exponents are already defined. Each stage involves a finite factorization and a finite union of finite supports. □

Definition 6.3 (Polynomial Pratt exponent of a monic polynomial)**.** Let
$$\mathcal{M} = \{f \in \mathbb{Q}[x] : f \text{ is monic}\},$$
where $1 \in \mathcal{M}$. For $f \in \mathcal{M}$ and $\psi \in \mathcal{I}$, define
$$M_\psi(f) = \sum_{\varphi\in\mathcal{I}} v_\varphi(f) M_\psi(\varphi). \tag{8}$$
In particular, $M_\psi(1) = 0$.

Geometrically, one starts with the irreducible factors of f, counted with multiplicity. A node labelled φ has as children the irreducible factors of $D\varphi$, again counted with multiplicity. Then $M_\psi(f)$ is the total number of nodes labelled ψ.

Proposition 6.4 (Complete additivity)**.** *For monic $f, g \in \mathbb{Q}[x]$ and $\psi \in \mathcal{I}$,*
$$M_\psi(fg) = M_\psi(f) + M_\psi(g).$$

Proof. This follows from
$$v_\varphi(fg) = v_\varphi(f) + v_\varphi(g)$$
and the definition (8). □

Example 6.5. Let $\varphi = x^2 + 1$. Then
$$D\varphi = x^2,$$
so its polynomial Pratt tree has one node labelled $x^2 + 1$ and two terminal nodes labelled x. Thus
$$M_{x^2+1}(x^2 + 1) = 1, \qquad M_x(x^2 + 1) = 2.$$
For
$$f = (x + 1)(x^2 + 1)$$
we have
$$M_{x+1}(f) = 1, \qquad M_{x^2+1}(f) = 1, \qquad M_x(f) = 3.$$

7 General reconstruction of monic polynomials

We now prove the main structural identity.

Theorem 7.1 (Polynomial Pratt reconstruction)**.** *For every monic polynomial $f \in \mathbb{Q}[x]$,*
$$\boxed{f(x) = \prod_{\varphi\in\mathcal{I}} \left(\frac{\varphi(x)}{D\varphi(x)}\right)^{M_\varphi(f)} = \prod_{\varphi\in\mathcal{I}} \Delta(\varphi)^{M_\varphi(f)}.} \tag{3}$$
Only finitely many factors differ from 1*.*

Proof. For a monic polynomial h, define the rational function

$$R(h) = \prod_{\psi \in \mathcal{I}} \Delta(\psi)^{M_\psi(h)}.$$

The product is finite by Theorem 6.2. Complete additivity implies

$$R(h_1 h_2) = R(h_1)R(h_2).$$

It is therefore enough to prove $R(\varphi) = \varphi$ for every $\varphi \in \mathcal{I}$.

For $\varphi = x$, the only nonzero coordinate is $M_x(x) = 1$, and

$$R(x) = \Delta(x) = \frac{x}{1} = x.$$

If φ is linear and $\varphi \neq x$, then $D\varphi = x$. By the defining recursion,

$$R(\varphi) = \Delta(\varphi)R(D\varphi) = \frac{\varphi}{x}x = \varphi.$$

Now assume by induction that $R(\eta) = \eta$ for all irreducibles of degree less than d, and let $\deg \varphi = d \geq 2$. Equation (7) gives

$$\begin{aligned} R(\varphi) &= \Delta(\varphi) \prod_{\psi \in \mathcal{I}} \Delta(\psi)^{M_\psi(D\varphi)} \\ &= \Delta(\varphi)R(D\varphi). \end{aligned}$$

All irreducible factors of $D\varphi$ have degree less than d, so multiplicativity and the induction hypothesis yield $R(D\varphi) = D\varphi$. Consequently,

$$R(\varphi) = \frac{\varphi}{D\varphi} D\varphi = \varphi.$$

Finally, factor a general monic polynomial as $f = \prod \varphi^{v_\varphi(f)}$. Multiplicativity gives

$$R(f) = \prod_{\varphi} R(\varphi)^{v_\varphi(f)} = \prod_{\varphi} \varphi^{v_\varphi(f)} = f.$$

$\square$

Remark 7.2 (Telescope rather than termwise polynomiality)**.** The individual factors $\Delta(\varphi) = \varphi/D\varphi$ are usually rational functions, not polynomials. The theorem says that the product prescribed by the consistent Pratt coordinates telescopes to a polynomial. Arbitrary finitely supported vectors in $\mathbb{N}_0^{(\mathcal{I})}$ need not have this property.

Example 7.3. For $f = (x+1)(x^2+1)$, Theorem 6.5 gives

$$\begin{aligned} \prod_{\varphi} \Delta(\varphi)^{M_\varphi(f)} &= \left(\frac{x}{1}\right)^3 \left(\frac{x+1}{x}\right) \left(\frac{x^2+1}{x^2}\right) \\ &= (x+1)(x^2+1). \end{aligned}$$

7.1 Extension to nonmonic polynomials

Although the main Hilbert-space discussion will remain monic, the scalar can be restored separately.

Corollary 7.4 (General nonzero rational polynomial)**.** *Every nonzero $f \in \mathbb{Q}[x]$ has a unique representation*

$$f = \lambda f_0, \qquad \lambda = \mathrm{lc}(f) \in \mathbb{Q}^\times, \qquad f_0 \in \mathcal{M}.$$

Then

$$f(x) = \lambda \prod_{\varphi \in \mathcal{I}} \Delta(\varphi)^{M_\varphi(f_0)}.$$

Thus the pair

$$(\lambda, (M_\varphi(f_0))_{\varphi \in \mathcal{I}})$$

reconstructs f.

8 Specialization to f_n: formula (3) implies formula (2)

Lemma 8.1 (The predecessor of f_p)**.** *For every prime p,*

$$Df_p = f_{p-1}.$$

Proof. For $p = 2$, this is the convention $Df_2 = Dx = 1 = f_1$. Let p be odd. The defining recursion gives

$$f_p = 1 + f_{p-1}.$$

Since $p - 1$ is even, Theorem 3.2 gives $f_{p-1}(0) = 0$, hence $f_p(0) = 1$. Therefore

$$Df_p = f_p - f_p(0) = f_p - 1 = f_{p-1}.$$

□

Because the f_p are pairwise distinct monic irreducibles, they are legitimate coordinate labels in $\mathcal{I}$.

Theorem 8.2 (Agreement of numerical and polynomial Pratt exponents)**.** *For all primes p and all positive integers n,*

$$M_{f_p}(f_n) = m_p(n). \tag{9}$$

Moreover, if $\varphi \in \mathcal{I}$ is not equal to f_p for any prime p, then

$$M_\varphi(f_n) = 0.$$

Proof. Both sides of (9) are completely additive in n: for the left side use $f_{ab} = f_a f_b$ and Theorem 6.4; for the right side use (2). It is therefore enough to compare them on a prime q. By the polynomial recursion and Theorem 8.1,

$$\begin{aligned} M_{f_p}(f_q) &= \mathbf{1}_{\{p=q\}} + M_{f_p}(Df_q) \\ &= \mathbf{1}_{\{p=q\}} + M_{f_p}(f_{q-1}), \end{aligned}$$

which is exactly the numerical recursion

$$m_p(q) = \mathbf{1}_{\{p=q\}} + m_p(q-1).$$

Strong induction on q proves equality.

For the final statement, every node descending from f_q is an irreducible factor of some f_{r-1}. By definition,

$$f_{r-1} = \prod_{s|r-1} f_s^{v_s(r-1)},$$

so all labels remain inside the family $\{f_p : p \in \mathbb{P}\}$. □

Theorem 8.3 (Polynomial Pratt product for f_n)**.** *For every $n \geq 1$,*

$$\boxed{f_n(x) = \prod_{p\in\mathbb{P}} \left(\frac{f_p(x)}{f_{p-1}(x)}\right)^{m_p(n)} = \prod_{p\in\mathbb{P}} \left(\frac{f_{p-1}(x)}{f_p(x)}\right)^{-m_p(n)}.} \tag{2}$$

Proof. Apply the general reconstruction formula (3) to f_n. By Theorem 8.2, only coordinates indexed by f_p occur, and their exponents are $m_p(n)$. By Theorem 8.1,

$$\Delta(f_p) = \frac{f_p}{Df_p} = \frac{f_p}{f_{p-1}}.$$

Substitution gives formula (2). □

Thus formula (2) is not an independent identity: it is the restriction of formula (3) to the recursively closed irreducible family $\{f_p\}$.

9 Evaluation at $x = 2$: formula (2) implies formula (1)

Theorem 9.1 (Numerical Pratt product)**.** *For every $n \geq 1$,*

$$n = \prod_{p\in\mathbb{P}} \left(1 - \frac{1}{p}\right)^{-m_p(n)}.$$

Proof. Evaluate formula (2) at $x = 2$. By Theorem 3.2,

$$f_n(2) = n, \qquad f_p(2) = p, \qquad f_{p-1}(2) = p - 1.$$

Hence

$$\begin{aligned} n &= \prod_{p\in\mathbb{P}} \left(\frac{p}{p-1}\right)^{m_p(n)} \\ &= \prod_{p\in\mathbb{P}} \left(1 - \frac{1}{p}\right)^{-m_p(n)}. \end{aligned}$$

□

We have therefore established the announced chain

$$\boxed{(3) \Longrightarrow (2) \Longrightarrow (1).}$$

10 Reconstruction and Hilbert-space coordinates

10.1 Intrinsic indexing

Let

$$\ell^2(\mathcal{I}) = \left\{ a = (a_\varphi)_{\varphi\in\mathcal{I}} : \sum_{\varphi\in\mathcal{I}} |a_\varphi|^2 < \infty \right\}.$$

The finitely supported vectors form the dense subspace $c_{00}(\mathcal{I})$.

Definition 10.1 (Polynomial Pratt feature map)**.** Define

$$\Phi_{\mathrm{poly}} : \mathcal{M} \longrightarrow \ell^2(\mathcal{I}), \qquad \Phi_{\mathrm{poly}}(f) = \big(M_\varphi(f)\big)_{\varphi\in\mathcal{I}}.$$

Theorem 10.2 (Hilbert-space embedding of monic polynomials)**.** *The map Φ_{poly} has the following properties:*

(i) $\Phi_{\mathrm{poly}}(f) \in c_{00}(\mathcal{I}; \mathbb{N}_0)$ *for every f;*

(ii) $\Phi_{\mathrm{poly}}(fg) = \Phi_{\mathrm{poly}}(f) + \Phi_{\mathrm{poly}}(g)$*;*

(iii) Φ_{poly} *is injective.*

Consequently, the multiplicative monoid of monic rational polynomials embeds into the additive cone of a real Hilbert space.

Proof. Finite support is Theorem 6.2; additivity is Theorem 6.4. If

$$\Phi_{\mathrm{poly}}(f) = \Phi_{\mathrm{poly}}(g),$$

then formula (3) reconstructs the same rational-function product from both vectors, so $f = g$. □

The induced kernel, norm, and metric are

$$K_{\mathrm{poly}}(f,g) = \langle \Phi_{\mathrm{poly}}(f), \Phi_{\mathrm{poly}}(g)\rangle = \sum_{\varphi\in\mathcal{I}} M_\varphi(f) M_\varphi(g), \tag{10}$$

$$\|f\|_{\mathrm{Pratt}} = \left(\sum_{\varphi\in\mathcal{I}} M_\varphi(f)^2 \right)^{1/2}, \tag{11}$$

$$d_{\mathrm{Pratt}}(f,g) = \left(\sum_{\varphi\in\mathcal{I}} \big(M_\varphi(f) - M_\varphi(g)\big)^2 \right)^{1/2}. \tag{12}$$

The notation $\|f\|_{\mathrm{Pratt}}$ is a norm on the embedded vectors, not a vector-space norm on $\mathcal{M}$, since $\mathcal{M}$ is not a vector space under multiplication.

Remark 10.3 (The image is not the whole lattice)**.** The image is a proper additive submonoid of $c_{00}(\mathcal{I}; \mathbb{N}_0)$. For example, the isolated vector having coordinate 1 at $x+1$ and zero elsewhere would reconstruct to

$$\frac{x+1}{x},$$

not to a polynomial. Valid coordinate vectors satisfy recursive consistency conditions encoded by the polynomial Pratt forests.

10.2 A concrete enumeration of the irreducibles

No enumeration is needed for the intrinsic space $\ell^2(\mathcal{I})$. To identify it with the standard space $\ell^2(\mathbb{N})$, however, one may choose an explicit ordering.

For a monic polynomial

$$\varphi = x^d + a_{d-1}x^{d-1} + \cdots + a_0 \in \mathbb{Q}[x],$$

clear denominators and divide by the gcd of the integer coefficients. This gives a unique primitive integer polynomial

$$F_\varphi = A_d x^d + A_{d-1}x^{d-1} + \cdots + A_0,$$

with $A_d > 0$ and

$$\varphi = \frac{F_\varphi}{A_d}.$$

Define the primitive height

$$H_{\mathrm{prim}}(\varphi) = \max_{0 \le j \le d} |A_j|$$

and the complexity

$$C(\varphi) = \deg \varphi + H_{\mathrm{prim}}(\varphi).$$

For a fixed value of C, only finitely many primitive coefficient vectors occur. We may therefore order all monic rational polynomials by:

1. increasing $C(\varphi)$;
2. increasing degree;
3. lexicographic order of the bounded integer tuple $(A_d, A_{d-1}, \ldots, A_0)$.

Filtering this list to the irreducibles produces

$$\mathcal{I} = \{\varphi_1, \varphi_2, \varphi_3, \ldots\}.$$

One may place x first by convention. The feature map then becomes the ordinary sequence

$$\Phi_{\mathrm{poly}}(f) = \big(M_{\varphi_1}(f), M_{\varphi_2}(f), \ldots\big) \in \ell^2(\mathbb{N}).$$

Different enumerations differ only by a permutation of coordinates, hence by a unitary operator on $\ell^2(\mathbb{N})$. The resulting geometry is therefore canonical up to unitary coordinate permutation.

10.3 The integer Hilbert-space embedding

Similarly, enumerate the primes increasingly and define

$$\Phi_{\mathrm{int}} : \mathbb{N} \longrightarrow \ell^2(\mathbb{P}), \qquad \Phi_{\mathrm{int}}(n) = \big(m_p(n)\big)_{p \in \mathbb{P}}.$$

Every vector has finite support, and

$$\Phi_{\mathrm{int}}(ab) = \Phi_{\mathrm{int}}(a) + \Phi_{\mathrm{int}}(b).$$

Formula (1) proves injectivity. Thus natural numbers and monic rational polynomials admit parallel sparse Hilbert-space coordinate systems.

For integers there is also a fixed linear readout of the logarithm. Put

$$w_p = -\log\left(1 - \frac{1}{p}\right).$$

Then

$$\log n = \sum_p w_p m_p(n) = \langle \Phi_{\mathrm{int}}(n), w \rangle ,$$

and $w \in \ell^2(\mathbb{P})$ because $w_p \sim 1/p$. A direct polynomial analogue would require choosing a suitable scalar-valued functional on $\mathbb{Q}(x)^\times$; evaluation at $x = 2$ recovers precisely the integer readout on the subfamily f_n.

11 A first dictionary of analogies

The following table is intended to grow in later versions.

Concept	**Positive integers**	**Monic polynomials in $\mathbb{Q}[x]$**
Ambient monoid	$(\mathbb{N}, \cdot)$ with unit 1	$(\mathcal{M}, \cdot)$ with unit 1
Atoms	primes p	monic irreducibles φ
Ordinary coordinates	$v_p(n)$	$v_\varphi(f)$
Recursive predecessor	$p \mapsto p - 1$	$\varphi \mapsto D\varphi$
Terminal atom	2, because $2 - 1 = 1$	x, with the convention $Dx = 1$
Recursive labels	prime factors of $p - 1$	irreducible factors of $D\varphi$
Pratt coordinates	$m_p(n)$	$M_\varphi(f)$
Complete additivity	$m_p(ab) = m_p(a) + m_p(b)$	$M_\varphi(fg) = M_\varphi(f) + M_\varphi(g)$
Local reconstruction weight	$p/(p-1)$	$\varphi/(D\varphi)$
Global reconstruction	$n = \prod_p (p/(p-1))^{m_p(n)}$	$f = \prod_\varphi (\varphi/D\varphi)^{M_\varphi(f)}$
Sparse feature vector	$(m_p(n))_p \in c_{00}(\mathbb{P})$	$(M_\varphi(f))_\varphi \in c_{00}(\mathcal{I})$
Hilbert space	$\ell^2(\mathbb{P}) \cong \ell^2(\mathbb{N})$	$\ell^2(\mathcal{I}) \cong \ell^2(\mathbb{N})$
Multiplication becomes	vector addition	vector addition
Special bridge	n	f_n, with $f_n(2) = n$

Concept	Positive integers	Monic polynomials in $\mathbb{Q}[x]$
Primality / irreducibility	n is prime	f_n is irreducible
Coordinate consistency	not every arbitrary vector need be a Pratt vector of an integer	not every arbitrary vector telescopes to a polynomial

Remark 11.1 (Ordinary factorization already embeds)**.** The ordinary valuation vectors $(v_p(n))_p$ and $(v_\varphi(f))_\varphi$ also give injective finite-support embeddings. The purpose of Pratt coordinates is not merely to prove that an embedding exists. Their additional content is recursive ancestry: a top-level atom contributes coordinates from all predecessors below it. This changes the induced geometry and creates a direct bridge between $p-1$ and $D\varphi$.

Part II

Mason–Stothers in Pratt Coordinates

12 The Pratt boundary and ordinary factorization

The reconstruction formula contains more information than injectivity. It also permits the ordinary irreducible exponents to be recovered by a linear "boundary" operation on the Pratt coordinates. In this part, the Pratt coordinates of a nonzero, not necessarily monic polynomial f mean those of its monic normalization

$$\widehat{f} = \frac{f}{\mathrm{lc}(f)}.$$

Thus multiplication by a nonzero scalar does not change any $M_\varphi(f)$.

Definition 12.1 (Pratt boundary coefficient)**.** For $0 \neq f \in \mathbb{Q}[x]$ and $\theta \in \mathcal{I}$, define

$$\beta_\theta(f) := M_\theta(f) - \sum_{\varphi \in \mathcal{I}} M_\varphi(f) v_\theta(D\varphi). \tag{13}$$

The sum is finite.

Theorem 12.2 (Boundary inversion)**.** *For every $0 \neq f \in \mathbb{Q}[x]$ and every $\theta \in \mathcal{I}$,*

$$\boxed{\beta_\theta(f) = v_\theta(f).}$$

Proof. Apply the ordinary valuation v_θ to the Pratt reconstruction formula for $\widehat{f}$:

$$\widehat{f} = \prod_{\varphi \in \mathcal{I}} \left(\frac{\varphi}{D\varphi}\right)^{M_\varphi(f)}.$$

Since $v_\theta(\varphi) = \mathbf{1}_{\{\theta=\varphi\}}$, one obtains

$$\begin{aligned} v_\theta(f) &= \sum_{\varphi \in \mathcal{I}} M_\varphi(f)\big(v_\theta(\varphi) - v_\theta(D\varphi)\big) \\ &= M_\theta(f) - \sum_{\varphi \in \mathcal{I}} M_\varphi(f) v_\theta(D\varphi) \\ &= \beta_\theta(f). \end{aligned}$$

□

The interpretation is useful. The coordinate $M_\theta(f)$ counts every occurrence of θ in the full Pratt forest. The second term in (13) subtracts occurrences that arise as children of higher nodes. What remains is precisely the multiplicity of θ among the roots of the forest, namely its ordinary multiplicity in f.

Definition 12.3 (Pratt radical)**.** For $0 \neq f \in \mathbb{Q}[x]$, define

$$\operatorname{rad}_P(f) := \prod_{\substack{\theta \in \mathcal{I} \\ \beta_\theta(f) > 0}} \theta.$$

Corollary 12.4 (Recovery of the ordinary radical)**.** *For every* $0 \neq f \in \mathbb{Q}[x]$,

$$\boxed{\operatorname{rad}_P(f) = \operatorname{rad}(f).}$$

Proof. By Theorem 12.2, the condition $\beta_\theta(f) > 0$ is equivalent to $v_\theta(f) > 0$. □

Example 12.5 (Internal nodes are removed by the boundary)**.** For $f = x^2 + 1$ one has

$$M_{x^2+1}(f) = 1, \qquad M_x(f) = 2.$$

However, x is not an ordinary factor of f. Since $D(x^2 + 1) = x^2$,

$$\beta_x(f) = 2 - 1 \cdot v_x(x^2) = 0, \qquad \beta_{x^2+1}(f) = 1.$$

Thus $\operatorname{rad}_P(f) = x^2 + 1$.

13 Logarithmic differentiation of the Pratt reconstruction

The ordinary derivative will be denoted by a prime; it must not be confused with the predecessor map D.

Theorem 13.1 (Pratt logarithmic derivative)**.** *For every* $0 \neq f \in \mathbb{Q}[x]$,

$$\frac{f'}{f} = \sum_{\varphi \in \mathcal{I}} M_\varphi(f) \left(\frac{\varphi'}{\varphi} - \frac{(D\varphi)'}{D\varphi} \right) \tag{14}$$

$$= \sum_{\theta \in \mathcal{I}} \beta_\theta(f) \frac{\theta'}{\theta}. \tag{15}$$

Proof. Take the logarithmic derivative of the reconstruction formula. This gives (14). Factor each $D\varphi$ into monic irreducibles and use

$$\frac{(D\varphi)'}{D\varphi} = \sum_{\theta \in \mathcal{I}} v_\theta(D\varphi) \frac{\theta'}{\theta}.$$

Collecting the coefficient of θ'/θ gives precisely $\beta_\theta(f)$. □

Corollary 13.2 (Pratt derivative divisibility)**.** *For every* $0 \neq f \in \mathbb{Q}[x]$*,*

$$\boxed{\frac{f}{\operatorname{rad}_P(f)} \mid f'.}$$

Equivalently,

$$\operatorname{rad}_P(f)\frac{f'}{f} \in \mathbb{Q}[x].$$

Proof. Multiplying (15) by $\operatorname{rad}_P(f)$ clears all denominators:

$$\operatorname{rad}_P(f)\frac{f'}{f} = \sum_{\beta_\theta(f)>0} \beta_\theta(f)\theta' \prod_{\substack{\eta\neq\theta \\ \beta_\eta(f)>0}} \eta.$$

The right-hand side is a polynomial. □

This is the point at which the Pratt reconstruction enters the Mason–Stothers proof. In a conventional proof, the divisibility is read directly from ordinary factorization. Here it is obtained by reconstruction, boundary inversion, and logarithmic differentiation.

14 The Mason–Stothers theorem in Pratt form

Theorem 14.1 (Pratt–Mason–Stothers)**.** *Let* $a, b, c \in \mathbb{Q}[x] \setminus \{0\}$ *be pairwise coprime and satisfy*

$$a + b = c.$$

Assume that they are not all constant. Then

$$\boxed{\max\{\deg a, \deg b, \deg c\} \leq \deg \operatorname{rad}_P(abc) - 1.} \tag{16}$$

Proof. Set

$$W = ab' - a'b.$$

Using $c = a + b$, one checks that the same polynomial can be written as

$$W = ac' - a'c = cb' - c'b.$$

By Theorem 13.2,

$$\frac{a}{\operatorname{rad}_P(a)} \mid a', \qquad \frac{b}{\operatorname{rad}_P(b)} \mid b', \qquad \frac{c}{\operatorname{rad}_P(c)} \mid c'.$$

It follows from the three displayed forms of W that

$$\frac{a}{\operatorname{rad}_P(a)}, \quad \frac{b}{\operatorname{rad}_P(b)}, \quad \frac{c}{\operatorname{rad}_P(c)}$$

all divide W. They are pairwise coprime, because a, b, c are pairwise coprime. Therefore

$$\frac{abc}{\operatorname{rad}_P(abc)} \mid W. \tag{17}$$

The Wronskian W is nonzero. Indeed, $W = 0$ would imply $(a/b)' = 0$, hence $a/b \in \mathbb{Q}$ in characteristic zero. Pairwise coprimality would then force a and b, and consequently c, to be constant.

Taking degrees in (17) gives

$$\deg a + \deg b + \deg c - \deg \operatorname{rad}_P(abc) \le \deg W.$$

Since

$$\deg W \le \deg a + \deg b - 1,$$

we obtain

$$\deg c \le \deg \operatorname{rad}_P(abc) - 1.$$

The other two inequalities follow from the other two forms of W. □

By Theorem 12.4, this is the classical Mason–Stothers theorem. The proof is nevertheless organized through the polynomial Pratt reconstruction. The classical theorem and its Wronskian proof go back to Stothers, Mason, and Snyder; see [6, 7, 8, 9].

15 Degree is the terminal Pratt coordinate

Theorem 15.1 (The x-coordinate theorem)**.** *For every* $0 \neq f \in \mathbb{Q}[x]$*,*

$$\boxed{M_x(f) = \deg f.}$$

In particular, for every $\varphi \in \mathcal{I}$*,*

$$M_x(\varphi) = \deg \varphi.$$

Proof. Take degrees at infinity in the reconstruction formula. For $\varphi \neq x$, the predecessor $D\varphi = \varphi - \varphi(0)$ has the same degree as φ, so

$$\deg \varphi - \deg D\varphi = 0.$$

For $\varphi = x$, one has $Dx = 1$, and the difference is 1. Therefore

$$\deg f = \sum_{\varphi \in \mathcal{I}} M_\varphi(f)(\deg \varphi - \deg D\varphi) = M_x(f).$$

□

The Mason–Stothers theorem can now be written entirely in Pratt coordinates.

Corollary 15.2 (Pure Pratt-coordinate form)**.** *Under the hypotheses of Theorem 14.1,*

$$\boxed{\max\{M_x(a), M_x(b), M_x(c)\} \le \sum_{\substack{\theta \in \mathcal{I} \\ \beta_\theta(abc) > 0}} M_x(\theta) - 1.} \tag{18}$$

Thus the left side is the maximum number of terminal x-nodes in the three Pratt forests, while the right side is the total number of terminal x-nodes below the distinct boundary roots of the forest of abc, minus one.

16 Can the terminal coordinate be replaced by an arbitrary coordinate?

For $\psi \in \mathcal{I}$, define the ψ-mass of the Pratt radical by

$$R_\psi(f) := M_\psi(\mathrm{rad}_P(f)) = \sum_{\substack{\theta \in \mathcal{I} \\ \beta_\theta(f) > 0}} M_\psi(\theta). \tag{19}$$

The most direct proposed extension of (18) would be

$$\max\{M_\psi(a), M_\psi(b), M_\psi(c)\} \overset{?}{\leq} R_\psi(abc) - \mathbf{1}_{\{\psi = x\}}. \tag{20}$$

For $\psi = x$, this is exactly Mason–Stothers. For other coordinates it is false, even without subtracting 1.

Proposition 16.1 (Failure of the naive coordinatewise extension)**.** *The inequality*

$$\max\{M_\psi(a), M_\psi(b), M_\psi(c)\} \leq R_\psi(abc)$$

need not hold for an irreducible $\psi \neq x$.

Proof. Take

$$\psi = x + 1, \qquad a = (x+1)^2, \qquad b = 1, \qquad c = (x+1)^2 + 1 = x^2 + 2x + 2.$$

The three polynomials are pairwise coprime and $a + b = c$. The polynomial c is irreducible over $\mathbb{Q}$, since its discriminant is -4. Moreover,

$$Dc = x^2 + 2x = x(x+2),$$

and the Pratt descendants of c contain no node labelled $x + 1$. Hence

$$M_{x+1}(a) = 2, \qquad M_{x+1}(b) = M_{x+1}(c) = 0.$$

The distinct irreducible factors of abc are $x + 1$ and c, so

$$R_{x+1}(abc) = M_{x+1}(x+1) + M_{x+1}(c) = 1.$$

Thus the proposed inequality would say $2 \leq 1$. □

A quadratic-coordinate counterexample is equally simple. Let

$$\psi = x^2 + 1, \qquad a = \psi^2, \qquad b = 1, \qquad c = \psi^2 + 1 = x^4 + 2x^2 + 2.$$

The polynomial c is Eisenstein at 2. Since

$$Dc = x^4 + 2x^2 = x^2(x^2 + 2),$$

its Pratt tree contains no $x^2 + 1$. Consequently the left side is 2 and $R_\psi(abc) = 1$.

16.1 SymPy exploration

Before attempting a proof, the naive inequality was tested computationally. A reproducible sample of 3000 triples was generated as follows:

- a and b had independently chosen degrees between 0 and 4;
- their integer coefficients were chosen in $[-2, 2]$, with nonzero leading coefficient;
- $c = a + b$ was required to be nonzero and the triple not entirely constant;
- only pairs with $\gcd(a, b) = 1$ were retained;
- factorization and all Pratt descendants were computed over $\mathbb{Q}$ with SymPy;
- the pseudorandom seed was 0.

The number of failures of (20) was:

ψ	x	$x+1$	$x-1$	x^2+1
failures among 3000 triples	0	38	34	1

Targeted square examples such as those above produce failures for many further irreducibles, including $x^2 + x + 1$, $x^2 + 2$, and $x^3 + x + 1$.

The obstruction is structural. The Wronskian argument loses one unit of ordinary degree because differentiation lowers degree. By Theorem 15.1, ordinary degree is exactly the terminal x-coordinate. Differentiation has no analogous monotonic behavior for a general internal coordinate M_ψ.

17 A valid Mason bound for every Pratt coordinate

Although the direct substitution $x \rightsquigarrow \psi$ fails, every coordinate admits a universal bound through its terminal capacity.

Lemma 17.1 (Subtree capacity)**.** *For every $\psi \in \mathcal{I}$ and every $0 \neq f \in \mathbb{Q}[x]$,*

$$\boxed{M_x(\psi)M_\psi(f) \leq M_x(f).} \tag{21}$$

Equivalently,

$$(\deg \psi)M_\psi(f) \leq \deg f.$$

Proof. In the Pratt forest of f, every occurrence of a node labelled ψ is the root of a subtree containing exactly

$$M_x(\psi) = \deg \psi$$

terminal x-nodes. The subtrees rooted at distinct occurrences of ψ are disjoint, because degree strictly decreases below every nonterminal irreducible node. The whole forest contains

$$M_x(f) = \deg f$$

terminal x-nodes. Counting the terminal nodes below all ψ-occurrences proves the inequality. □

Corollary 17.2 (All-coordinate Pratt–Mason capacity bound)**.** *Under the hypotheses of Theorem 14.1, for every $\psi \in \mathcal{I}$,*

$$\boxed{M_x(\psi)\max\{M_\psi(a), M_\psi(b), M_\psi(c)\} \le \sum_{\substack{\theta\in\mathcal{I}\\ \beta_\theta(abc)>0}} M_x(\theta) - 1.} \tag{22}$$

Equivalently,

$$\max\{M_\psi(a), M_\psi(b), M_\psi(c)\} \le \left\lfloor \frac{\deg \operatorname{rad}_P(abc) - 1}{\deg \psi} \right\rfloor .$$

Proof. Apply Theorem 17.1 to each of a, b, c, take the maximum, and then apply Theorems 14.1 and 15.1. □

The exact amount by which the naive coordinatewise radical can fail is measured by a nonnegative slack. Define

$$s_\psi(\theta) := M_x(\theta) - M_x(\psi)M_\psi(\theta) \ge 0.$$

Then

$$\begin{aligned} \deg \operatorname{rad}_P(abc) &= \sum_{\beta_\theta(abc)>0} M_x(\theta) \\ &= M_x(\psi)R_\psi(abc) + \sum_{\beta_\theta(abc)>0} s_\psi(\theta). \end{aligned}$$

Thus (22) becomes

$$M_x(\psi)\max M_\psi \le M_x(\psi)R_\psi(abc) + \sum_{\beta_\theta(abc)>0} s_\psi(\theta) - 1. \tag{23}$$

For $\psi = x$, all slack terms vanish and the classical theorem is recovered exactly. For an internal coordinate, the slack is generally indispensable.

18 Specialization to the family f_n and an integer research direction

The polynomial theorem already gives unconditional inequalities for numerical Pratt coordinates, but an additional term records the failure of the map $n \mapsto f_n$ to preserve addition.

For $n \ge 1$, define the squarefree terminal weight

$$\rho_2(n) := \sum_{p\mid n} m_2(p).$$

Since the distinct irreducible factors of f_n are the f_p with $p \mid n$,

$$\deg \operatorname{rad}(f_n) = \rho_2(n).$$

Theorem 18.1 (A Pratt–Mason inequality for two natural numbers)**.** *Let $a, b \in \mathbb{N}$ be coprime and put*

$$G_{a,b}(x) := f_a(x) + f_b(x).$$

Then

$$\boxed{\max\{m_2(a), m_2(b)\} \leq \rho_2(ab) + \deg \operatorname{rad}(G_{a,b}) - 1.} \tag{24}$$

More generally, for every prime p,

$$\boxed{m_2(p) \max\{m_p(a), m_p(b)\} \leq \rho_2(ab) + \deg \operatorname{rad}(G_{a,b}) - 1.} \tag{25}$$

Proof. The polynomials f_a and f_b are coprime because a and b have no common prime divisor. Hence $f_a, f_b, G_{a,b}$ are pairwise coprime and satisfy

$$f_a + f_b = G_{a,b}.$$

Apply Theorem 14.1. Since

$$\deg f_n = M_x(f_n) = m_2(n)$$

and

$$\deg \operatorname{rad}(f_a f_b) = \rho_2(ab),$$

we obtain (24). Applying Theorem 17.2 with $\psi = f_p$, and using

$$M_{f_p}(f_n) = m_p(n), \qquad M_x(f_p) = m_2(p),$$

gives (25). □

If $a + b = c$, then $G_{a,b}(2) = c$. Nevertheless, $G_{a,b}$ is generally not equal to f_c. The precise addition defect is

$$E_{a,b}(x) := \frac{f_a(x) + f_b(x) - f_{a+b}(x)}{x - 2} \in \mathbb{Z}[x], \tag{26}$$

because the numerator vanishes at $x = 2$. Thus

$$f_a + f_b = f_{a+b} + (x - 2)E_{a,b}.$$

Any attempt to derive an integer *abc* statement from the polynomial theorem must control this defect, or equivalently control the new irreducible factors in $f_a + f_b$.

This obstruction cannot be removed by choosing a different nonconstant encoding that preserves both operations.

Proposition 18.2 (No nonconstant semiring embedding)**.** *Let $F : \mathbb{N} \to \mathbb{Q}[x]$ satisfy*

$$F(1) = 1, \qquad F(m + n) = F(m) + F(n), \qquad F(mn) = F(m)F(n).$$

Then $F(n) = n$ is the constant polynomial for every n.

Proof. Additivity gives

$$F(n) = \underbrace{F(1) + \cdots + F(1)}_{n \text{ times}} = n.$$

□

Thus a nontrivial multiplicative polynomial model of the natural numbers must pay for addition by an error term. The "elephant in the room" is therefore not the Mason–Stothers step, which transfers cleanly, but the arithmetic problem of bounding the addition radical

$$\mathcal{A}(a,b) := \deg \operatorname{rad}(f_a + f_b)$$

or the Pratt complexity of $E_{a,b}$ in terms of $a, b, c = a + b$.

Preliminary computations show both promise and difficulty. For an integer polynomial G, write

$$\operatorname{Rad}_{\mathbb{Z}}(G) := |\operatorname{cont}(G)| \prod_i F_i,$$

where the $F_i \in \mathbb{Z}[x]$ are the distinct primitive irreducible factors, chosen with positive leading coefficient. Then

$$\bigl|\operatorname{Rad}_{\mathbb{Z}}(G)(2)\bigr| \mid |G(2)|.$$

For $G = G_{a,b}$ this divisor of $a + b$ need not be bounded by the ordinary integer radical of $a + b$. For example,

$$f_1 + f_3 = x + 2, \qquad \bigl|\operatorname{Rad}_{\mathbb{Z}}(x+2)(2)\bigr| = 4 > \operatorname{rad}(4) = 2,$$

and

$$\bigl|\operatorname{Rad}_{\mathbb{Z}}(f_2 + f_{23})(2)\bigr| = 25 > \operatorname{rad}(25) = 5.$$

In a survey of the 3044 coprime pairs $1 \le a \le b \le 100$, equality

$$\bigl|\operatorname{Rad}_{\mathbb{Z}}(f_a + f_b)(2)\bigr| = \operatorname{rad}(a+b)$$

held in 2176 cases, but the displayed examples show that it is not universal.

A plausible research program is therefore to study conditional or averaged bounds for one of the quantities

$$\deg \operatorname{rad}(f_a + f_b), \qquad \deg \operatorname{rad}(E_{a,b}), \qquad \Phi_{\text{poly}}(E_{a,b}),$$

and to determine whether special arithmetic classes of triples $a+b=c$ force these defects to be small. Such a result would yield a genuine, though modified, Pratt-coordinate analogue of an *abc* estimate. No such bound is claimed in the present version.

19 Questions for later versions

The following problems indicate possible directions for extending this note.

1. **Characterize the image.** Give necessary and sufficient conditions for a vector in $c_{00}(\mathcal{I}; \mathbb{N}_0)$ to equal $\Phi_{\text{poly}}(f)$ for some monic f.
2. **Transition matrix.** After ordering $\mathcal{I}$, compare the ordinary exponent vector

   $$v(f) = (v_\varphi(f))_\varphi$$

   with the Pratt vector

   $$M(f) = (M_\varphi(f))_\varphi.$$

   The relation $M(f) = Av(f)$ is governed by the columns $M(\varphi)$. Find orderings for which A has a useful triangular or block-triangular form.

3. **Polynomial Pratt order.** Define

$$f \preccurlyeq_P g \quad \Longleftrightarrow \quad M_\varphi(f) \le M_\varphi(g) \quad \text{for every } \varphi \in \mathcal{I}.$$

   Study meets, joins, local finiteness, and incidence algebras.

4. **Kernels.** Investigate kernels built from minima of polynomial Pratt coordinates, in analogy with gcd and Pratt-meet kernels for integers.

5. **Evaluation maps.** For which rational or complex values a does evaluation $f \mapsto f(a)$ convert polynomial Pratt reconstruction into a useful numerical product? What conditions avoid zeros and poles?

6. **Degree and height statistics.** Study how the Pratt norm and tree depth depend on the degree and primitive height of an irreducible polynomial.

7. **Galois information.** Determine whether the geometry of the vectors $\Phi_{\text{poly}}(f_p)$ reflects properties of splitting fields or Galois groups.

8. **Computation.** Enumerate monic irreducibles by degree and primitive height, compute their polynomial Pratt vectors, and search for clusters, extremal examples, and repeated local structures.

9. **Coordinate slack.** Study

$$s_\psi(\theta) = M_x(\theta) - M_x(\psi)M_\psi(\theta)$$

   and determine when the all-coordinate Mason bound is sharp.

10. **Addition defect for f_n.** Investigate the factorization and Pratt coordinates of

$$E_{a,b} = \frac{f_a + f_b - f_{a+b}}{x-2},$$

   and seek arithmetic conditions under which its radical complexity is controlled by the Pratt data of abc.

20 Conclusion

The predecessor map

$$D\varphi = \varphi - \varphi(0),$$

with the terminal convention $Dx = 1$, turns factorization of monic irreducible polynomials into a finite recursive forest. The label counts $M_\varphi(f)$ are completely additive, and the local ratios $\varphi/D\varphi$ telescope to reconstruct every monic polynomial. The family f_n is closed under the recursion, the irreducibles f_p reproduce numerical Pratt trees, and evaluation at 2 connects the polynomial and integer product formulas. Thus natural numbers and monic rational polynomials admit parallel sparse Hilbert-space coordinate systems.

The new application in Part II shows that ordinary factorization is the boundary of the recursive Pratt forest. This boundary recovers the radical, logarithmic differentiation gives the derivative divisibility needed in the Wronskian argument, and Mason–Stothers becomes a statement entirely about Pratt coordinates. The terminal coordinate is distinguished by the exact identity $M_x(f) = \deg f$. A direct replacement by an arbitrary internal coordinate is false, but every coordinate satisfies a sharp terminal-capacity constraint and therefore inherits a valid Mason bound.

For the integer family f_n, the polynomial theorem produces inequalities involving the numerical Pratt exponents. The remaining difficulty is addition: multiplication is represented exactly, whereas $f_a + f_b$ is generally not f_{a+b}. The quotient $(f_a + f_b - f_{a+b})/(x-2)$ records this defect. Understanding its factorization and Pratt geometry is a concrete route toward modified, rather than classical, *abc*-type statements.

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